

measured-rag

A RAG system over clinical literature —
and the evidence for how well it works.

243 PubMed papers · 262 chunks · 57 hand-written test questions

13 configurations compared on quality and latency

208 tests · FastAPI · Docker · CI-gated evaluation

Includes the ablations where the design lost.

github.com/Rejeb1/measured-rag

What this project is

A language model does not know your documents. Asked about them, it invents plausible answers. **RAG** fixes that: search real documents first, then force the model to answer only from what was found. This system does it over 243 real medical papers — and, unusually, **measures how well it works**.

243

PubMed papers

262

searchable chunks

57

test questions

13

configs compared

208

tests

The pipeline, stage by stage

0

Ingest & chunk

Fetch abstracts from PubMed's API, cut them into ~512-token chunks.

Never split mid-sentence — a chunk starting mid-sentence is useless as a citation.

1

Two searches in parallel

BM25 (keyword) and a dense bi-encoder (meaning) each return their top 50.

They fail in opposite ways: BM25 nails SGLT2 and ICD-10; dense handles paraphrase.

2

Fuse the two lists (RRF)

Reciprocal Rank Fusion: $\text{score} = 1/(60+\text{rank})$ summed across lists.

Uses ranks, not scores — so unbounded BM25 scores never need converting to cosines.

3

Rerank

A cross-encoder reads query and passage together, keeps the best 5 of 50.

The bi-encoder never saw them together. Far more accurate, ~1000x slower.

4

Assemble context

Deduplicate, order best-first-and-last, cap at 3000 tokens.

Models attend most to the start and end of a context window.

5

Generate with citations

Answer only from these passages, cite every claim, or say INSUFFICIENT_CONTEXT.

Fabricated citations are dropped — a made-up reference must never count as grounding.

Why two different searches?

Dense retrieval smooths away exact identifiers — **SGLT2**, **NCT03057951**, **126 mg/dL**. BM25 catches those but misses paraphrase entirely: it cannot connect "stops the kidney reabsorbing sugar" to "blocks glucose reabsorption in the proximal tubule". The BM25 tokenizer here is hand-written so **ICD-10** survives whole and emits **icd + 10**.

The evaluation layer

Thousands of people build the pipeline on the previous page. Almost nobody measures one. This is what the project is actually about.

Two questions, measured separately — never blended

	How it is measured	Speed	Trust
Did it <i>find</i> the right chunk?	recall@k, MRR, NDCG — no AI involved	milliseconds	high
Did it <i>use</i> the chunk correctly?	LLM-judge with an explicit rubric	~4 hours	low

Why the split is the whole point

It lets you say “*search is 0.903 but the writer is ignoring its sources*” — a diagnosis you can act on. A single blended score cannot tell you which half to fix. It also has a practical payoff: the retrieval half is cheap enough to run on **every commit** in CI; the generation half is not.

The golden set — 57 hand-written questions

Type	n	What it tests
factoid	30	answerable from one chunk
multi-hop	8	needs two chunks from different papers
aggregation	5	“how many”, “list all” — needs several chunks
unanswerable	14	the answer is not in the corpus at all

The unanswerable questions matter most

They are the only way to tell a system that *knows things* from one that *says anything* — and almost nobody includes them. They are deliberately **close** to the corpus (metformin and eGFR, not astrophysics), so retrieval fires confidently and the generator genuinely has to decide to refuse. An obviously off-topic question would be refused trivially and prove nothing.

Every ablation row is a config change, never a code change

That is what makes rows comparable. “Hybrid + reranker” and “dense only” are the same code with different switches — so a difference in the numbers can only come from the thing the label names. A test asserts that disabling a stage changes nothing else.

What the measurements said

Including three results that contradict the design the project argues for. All figures below: 57 questions, 43 answerable, GTX 1650.

Configuration	Recall@10	NDCG@10	MRR	p95 latency
Dense only	0.909	0.833	0.821	18 ms
BM25 only	0.864	0.802	0.810	1 ms
Hybrid (RRF)	0.877	0.820	0.820	2 ms
Hybrid (normalized-sum)	0.877	0.823	0.821	3 ms
Hybrid + rerank	0.903	0.846	0.845	837 ms
Hybrid + rerank, top-20 pool	0.892	0.838	0.840	804 ms
Hybrid + rerank, top-100 pool	0.911	0.849	0.848	4,025 ms
Hybrid + rerank + query rewrite	0.903	0.846	0.845	35,190 ms

1 · Dense-only beat the full pipeline

0.909 vs **0.903** recall, at **18 ms** vs **837 ms** — better and 46x faster. The reranker does buy ranking quality (NDCG 0.820 → 0.846), but it is winning back ground that fusion gave away.

2 · Adding BM25 hurt

Hybrid scored **0.877** against dense-only's **0.909** — 3.2 points worse. RRF fuses on rank, so a consistently weaker retriever drags a stronger one down. The score-fusion control is a wash, so the loss is fusion *itself*, not the fusion method.

3 · Query rewriting cost 10.9 seconds for +0.001 NDCG

An 18x latency increase for noise. Delete it. This number is only trustworthy because a silent no-op behind it was found and fixed first — see the next page.

Chunking — bigger won, and “smart” lost

Chunking	Recall@10	NDCG@10	MRR	note
Chunk 256 (structure)	0.768	0.558	0.502	only 28 / 43 scored — not comparable
Chunk 512 (structure)	0.903	0.846	0.845	the default
Chunk 1024 (structure)	0.911	0.889	0.907	best NDCG and MRR
Chunk 512 (fixed-size)	0.915	0.864	0.873	best recall — beats structure-aware

Four bugs — all the same failure mode

Each was a component **degrading politely** while silently invalidating a measurement. In a system whose only product is trustworthy numbers, graceful degradation is a hazard.

#	The bug	Why it was dangerous
1	Ollama's default num_ctx 4096 regardless of the model	Long prompts returned empty or degenerate output. Cost three golden-set builds before it was isolated.
2	BM25 empty result set out-of-vocabulary query	Zero results means the system refuses for the wrong reason — abstention becomes trivially correct.
3	Query rewrite no-op no LLM attached	The row matched the baseline to three decimals — which is exactly what a genuine null result looks like.
4	Abstention detector exact string match	Wanted INSUFFICIENT_CONTEXT, model wrote INSUFFICIENT CONTEXT. Every real refusal counted as a failure.

The conclusion that had to be retracted

An earlier draft reported “12 of 14 unanswerable questions were answered anyway — genuine hallucination risk” and called it the most serious defect in the system. Reading the raw answer text disproved it: **9 were GPU garbage**, 3 were degenerate, 2 were correct refusals, and **zero were hallucinations**. The metric was technically correct and the conclusion was false. A metric computed over an unexamined field will confidently describe something that never happened.

Stated limitations

Limitation	Why it is disclosed
The golden set is biased toward dense search	Questions were paraphrased rather than copied, which under-represents the exact-identifier case BM25 exists for. So “hybrid hurts” is true <i>on this question set</i> , not universally.
28% of generated answers collapsed	Sustained load on a 4GB laptop GPU. Every generation figure is computed on the remaining ~72%, so none should be quoted until re-run on adequate hardware.
The judge is a 3B model	It errored on 35% of items and collapsed relevance onto three rubric anchors. Judge validation against human labels is built, and is the prerequisite for citing any generation number.

Why the negative results stay in

A table where every row improves is a table that was pruned. Reporting that a component you built did **not** help — with the number attached — is a stronger signal of engineering judgement than a clean sweep. The same applies to the retracted conclusion above: it is left in the write-up on purpose.

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